



SYDNEY BOYS HIGH
SCHOOL
MOORE PARK, SURRY HILLS

Yearly Examination 2011

Mathematics 2 Unit

General Instructions

- Working time – 90 Minutes
- Reading Time – 5 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
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- All necessary working should be shown in every question.
- Answer in simplest exact form unless otherwise instructed.

Total Marks – 64

- All Questions may be attempted.
- Section A (Q1 & Q2) and Section B (Q3 & Q4) should be handed up in separate examination booklets.
- .
- Full Marks may not be awarded for careless or poorly set out work.

Examiner – *P.R. Bigelow*

Section A
[Start a new booklet]

Question 1. (17 Marks)

(a) Write down the exact values of: **2**

(i) $\sin 45^\circ$

(ii) $\cos \frac{7\pi}{6}$

(b) The point $(4, a)$ lies on the line $4x - 3y - 7 = 0$. Find the value of a . **1**

(c) Solve the following quadratic equations: **4**

(i) $x^2 - x - 12 = 0$.

(ii) $2x^2 + 4x - 5 = 0$ (Leave answer in simplest surd form.)

(d) What is the domain of the function $y = \sqrt{2 - x}$? **1**

(e) State whether the following functions are ODD, EVEN, or NEITHER. **2**

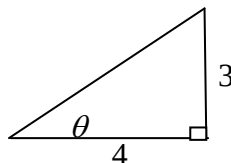
(i) $f(x) = x^2 + 12$

(ii) $f(x) = \frac{-x}{x^2 + 1}$

(f) Write down the value of: **2**

(i) $\cot \theta$

(ii) $\sin \theta$



Question 1 (Continued)

(g) If α and β are roots of $x^2 - 6x - 4 = 0$, find the values of:

5

(i) $\alpha + \beta$

(ii) $\alpha\beta$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta}$

(iv) $\alpha^2 + \beta^2$

Question 2. (16 Marks)

(a) What is the slope of the line $x + 3y - 5 = 0$? 1

(b) Sketch the following showing essential features: 4

(i) $y = x + 2$ (iii) $y = 2^x$

(ii) $xy = 2$ (iv) $y = |x - 2|$

(c) For the parabola $x^2 = 8(y - 1)$, write down the: 4

(i) co-ordinates of the vertex

(ii) focal length

(iii) co-ordinates of the focus

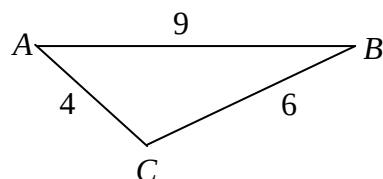
(iv) equation of the directrix.

(d) Find the equations of the following lines (giving your answer in general form). 4

(i) through $(4, 6)$ with slope -2 .

(ii) perpendicular to $x - 2y + 5 = 0$ and passing through $(3, 6)$.

(e) Find the size of the largest angle in the triangle (to the nearest minute). 3



Section B
[Start a new booklet]

Question 3. (15 Marks)

- (a) Find the value of θ (to the nearest degree) if **2**

$$\cos \theta = -0.7 \text{ and } 180^\circ \leq \theta \leq 360^\circ.$$

- (b) Show that $(\sin \phi + \cos \phi)^2 + (\sin \phi - \cos \phi)^2 = 2$. **2**

- (c) Differentiate $f(x) = x^2 - 6x + 1$ by first principles. **2**

- (d) For the parabola $y = x^2 - 6x + 1$, find: **3**

(i) the equation of the axis

(ii) the range of values of y .

- (e) Find the perpendicular distance of the point $(4,1)$ from the line $3x - 4y + 12 = 0$. **2**

- (f) Sketch the regions of the number plane defined by: **4**

(i) $x^2 + y^2 \leq 16$

(ii) $y > x^2 + 9$

Question 4 (16 Marks)

(a) Differentiate the following:

6

(i) $y = 3 - x + x^2$

(ii) $y = \sqrt{\frac{x}{x-2}}$

(iii) $y = \frac{x+1}{x-2}$

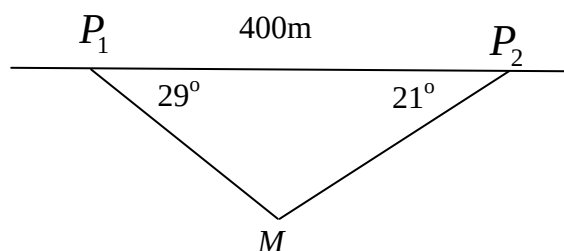
(iv) $y = (2 - 3x^2)^5$

(b) Find the centre and radius of the circle $x^2 - 4x + y^2 + 6y = 0$.

2

(c)

4



P_1 and P_2 are points 400m apart on horizontal ground. Mine shafts are driven from P_1 and P_2 to meet underground at M .

Find to the nearest metre

(i) the length of the shaft P_1M .

(ii) the vertical depth of M below the surface.

(d) For what values of k does $x^2 - (k+5)x + 9 = 0$ have

4

(i) equal roots.

(ii) no roots.

This is the end of the paper.

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QUESTION ONE

A) i) $\sin 45^\circ = \frac{1}{\sqrt{2}}$

ii) $\cos \frac{7\pi}{6} = \cos 210^\circ = +\cos(180+30) = -\cos 30 = -\frac{\sqrt{3}}{2}$

b) (A.a) $4x - 3y - 7 = 0$

SUB $x=4$ $y=a$ $4(4) - 3a - 7 = 0$

$$-3a + 9 = 0$$

$$a = 3 \quad \text{--- (1)}$$

c) i) $x^2 - x - 12 = 0$

$$(x-4)(x+3) = 0$$

$$x = 4 \text{ or } x = -3 \quad \text{--- (2)}$$

ii) $2x^2 - 4x - 5 = 0$

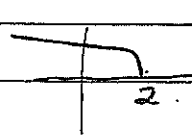
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{4 \pm \sqrt{16 - 4(2)(-5)}}{4}$$

$$= \frac{4 \pm \sqrt{56}}{4}$$

$$= \frac{4 \pm 2\sqrt{14}}{4}$$

$$= \frac{2 \pm \sqrt{14}}{2} \quad \text{--- (2)}$$

d) $y = \sqrt{2-x}$  --- (1)

domain $x \in \mathbb{R} : x \leq 2$

e) i) $f(x) = x^2 - 12$ --- (1) ii) $f(x) = \frac{-x}{x^2 + 1}$ --- (1)

$$f(-x) = (-x)^2 - 12 = x^2 - 12$$

$$f(-x) = \frac{-(-x)}{(-x)^2 + 1} = \frac{x}{x^2 + 1}$$

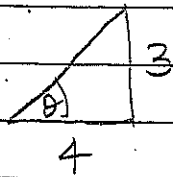
Since $f(x) = f(-x)$
fn is even

Since $f(x) = -f(-x)$
fn is odd

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Q1.

f):



$$i) \cot \theta = \frac{\text{adj}}{\text{opp}} = \frac{4}{3}$$

(1)

$$3^2 + 4^2 = 5^2$$

$$ii) \sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{3}{5}$$

(1)

$$g) x^2 - 6x - 4 = 0$$

$$i) \alpha + \beta = \frac{-b}{a} = +6$$

(1)

$$ii) \alpha\beta = \frac{c}{a} = -4$$

(1)

$$iii) \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha\beta} = \frac{b}{-4} = \frac{-3}{2}$$

(1)

$$\begin{aligned} iv) \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= (+6)^2 - 2(-4) \\ &= 36 + 8 \\ &= 44 \end{aligned}$$

(2)

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QUESTION TWO

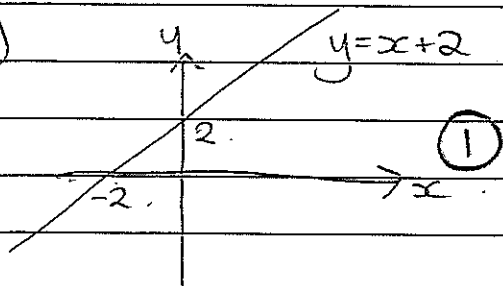
a) $x + 3y - 5 = 0$

$$3y = -x + 5$$

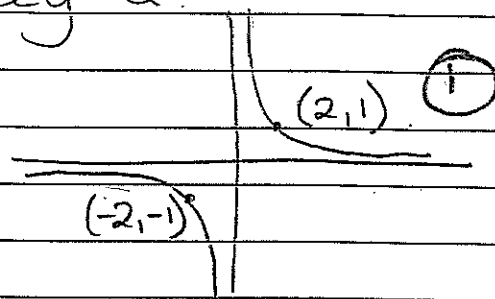
$$y = -\frac{1}{3}x + \frac{5}{3}$$

gradient = $-\frac{1}{3}$ (1)

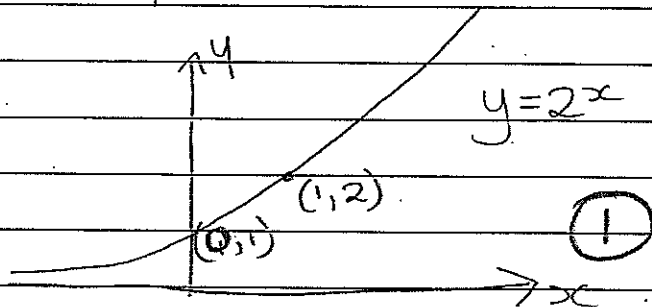
b) i)



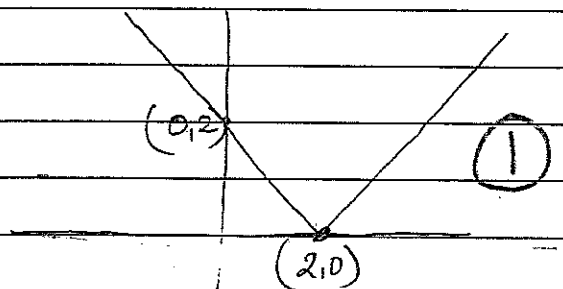
ii) $xy = 2$



iii)



iv) $y = |x - 2|$



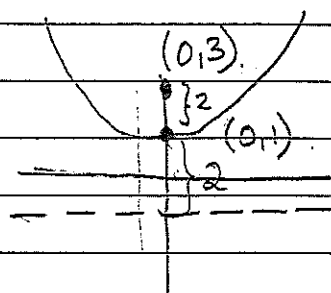
c) i) $x^2 = 8(y - 1)$ (1)

vertex (0, 1)

ii) focal length $a = 2$ (1)

iii) focus (0, 3) (1)

iv) $y = -1$ (1)



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QUESTION TWO.

d) i) $y - y_1 = m(x - x_1)$ (4,6) $m = -2$

$$y - 6 = -2(x - 4)$$

$$y - 6 = -2x + 8$$

$$2x + y - 14 = 0$$

— (2)

ii) $x - 2y + 5 = 0$

$$-2y = -x - 5$$

$$y = \frac{1}{2}x + \frac{5}{2}$$

grad = $\frac{1}{2}$
(3,6)

grad of perp = -2

$$-2 \times \frac{1}{2} = -1 \checkmark$$

$$y - 6 = -2(x - 3)$$

$$y - 6 = -2x + 6$$

$$2x + y - 12 = 0$$

— (2)

e) $\cos C = \frac{a^2 + b^2 - c^2}{2ab} =$

$$= \frac{6^2 + 4^2 - 9^2}{2(6)(4)}$$

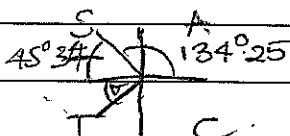
$$= 0.604 \text{ (3dp)}$$

$$C = 127^\circ 10'$$

— (3)

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QUESTION THREE

a) $\cos \theta = -0.7$ 

$$\theta = 134.25^\circ$$

$$\theta = 226^\circ$$

— (2)

b) $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2 = 2$ prove.

$$\begin{aligned} \text{LHS} &= \sin^2 \theta + 2\sin \theta \cos \theta + \cos^2 \theta + (\sin^2 \theta - 2\cos \theta \sin \theta + \cos^2 \theta) \\ &= (\sin^2 \theta + \cos^2 \theta) + (\sin^2 \theta + \cos^2 \theta) \end{aligned}$$

$$= 1 + 1$$

$$= 2$$

$$= \text{RHS}$$

— (2)

b) $f(x) = x^2 - bx + 1$

$$f(x+h) = (x+h)^2 - b(x+h) + 1$$

$$= x^2 + 2xh + h^2 - bx - bh + 1$$

$$= \cancel{x^2} + 2xh + h^2 - \cancel{bx} - bh + 1$$

$$f'(x) = \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right)$$

$$= \lim_{h \rightarrow 0} \left[\frac{(\cancel{x^2} + 2xh + h^2 - \cancel{bx} - bh + 1) - (\cancel{x^2} - \cancel{bx} + 1)}{h} \right]$$

$$= \lim_{h \rightarrow 0} \left(\frac{2xh + h^2 - bh}{h} \right)$$

$$= \lim_{h \rightarrow 0} (2x + h - b)$$

$$= 2x - b$$

— (2)

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QUESTION THREE.

d) $y = x^2 - 6x + 1$

i) $x = \frac{-b}{2a} = \frac{6}{2} = 3$ — (1)

ii) y value when $x = 3$

$$y = (3)^2 - 6(3) + 1$$

$$y = 9 - (18) + 1$$

$$= -9 + 1$$

$$= \underline{\underline{-8}}$$

Range $y \in \mathbb{R}: y \geq -8$ — (2)

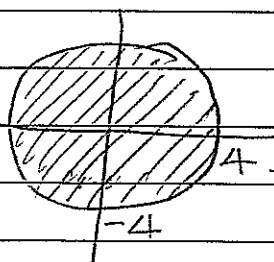
e) (x_1, y_1) $3x + 4y + 12 = 0$ $\text{hd} = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

$$\text{hd} = \frac{3(4) - 4(1) + 12}{\sqrt{3^2 + (-4)^2}}$$

$$= \frac{12 - 4 + 12}{5}$$

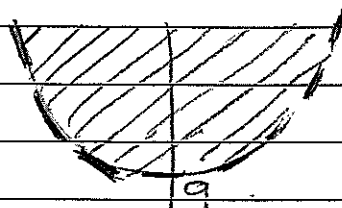
$$= 4 \text{ units} \text{ — (2)}$$

f) i) $x^2 + y^2 \leq 16$



(2)

ii) $y > x^2 + 9$



(2)

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QUESTION FOUR

a) i) $y = (3 - x + x^2)$

$$\frac{dy}{dx} = -1 + 2x \quad \text{--- (1)}$$

ii) $y = \sqrt{x} = x^{1/2}$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}} \quad \text{--- (1)}$$

iii) $y = \frac{x+1}{x-2}$ $u = x+1$ $u' = 1$
 $v = x-2$ $v' = 1$

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2} = \frac{(x-2)(1) - (x+1)(1)}{(x-2)^2}$$

$$= \frac{x-2-x+1}{(x-2)^2}$$

$$= \frac{-3}{(x-2)^2} \quad \text{--- (2)}$$

$$\text{iv) } y = (2 - 3x^2)^5$$
$$\frac{dy}{dx} = 5(2 - 3x^2)^4 \times -6x$$

$$= -30x(2-3x^2)^4 \quad \text{--- (2)}$$

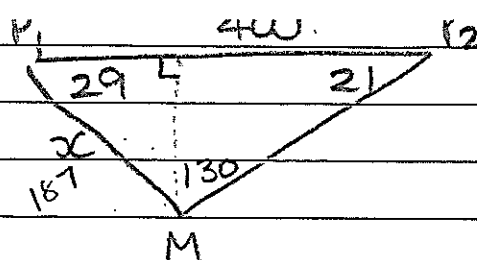
b) $x^2 - 4x + y^2 + 6y = 0$

$$x^2 - 4x + 4 + y^2 + 6y + 9 = 13$$

$$(x-2)^2 + (y+3)^2 = 13$$

Centre = $(2, -3)$ radius $\sqrt{13}$ — (2)

c)



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Q4.

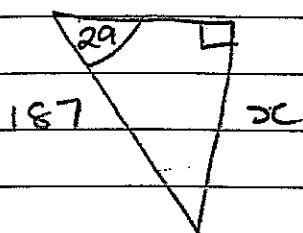
$P, M \Rightarrow$

$$\frac{x}{\sin 21} = \frac{400}{\sin 130}$$

$$x = \frac{400(\sin 21)}{\sin 130}$$

$$= 187 \text{ m (nearest m)}$$

— (2)



$$\sin 29 = \frac{x}{187}$$

$$x = 187 \sin 29$$

$$x = 90.72$$

$$= 91 \text{ m}$$

— (2)

d) $x^2 - (k+5)x + 9 = 0$

Equal roots $b^2 - 4ac = 0$

$$(-(k+5))^2 - 4(1)(9) = 0$$

$$k^2 + 10k + 25 - 36 = 0$$

$$k^2 + 10k - 11 = 0$$

$$(k+11)(k-1) = 0$$

$$k = -11 \text{ or } +1$$

No roots:

$$b^2 - 4ac < 0$$

$$(-(k+5))^2 - 4(1)(9) < 0$$

$$k^2 + 10k - 11 < 0$$

$$-11 < k < 1$$

